



SANKALP LAXMAN KHATAKE

AI/ML & Generative AI Engineer

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LinkedIn

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Portfolio

Baramati, Pune - 413133



CAREER PROFILE

AI/ML & Generative AI Intern skilled in LLMs, RAG, Agentic AI, Computer Vision, and MLOps. Proficient in Python, TensorFlow, LangChain, Docker, Kubernetes, AWS, and modern web frameworks. Hands-on experience with production-grade AI solutions, real-time inference pipelines, and cloud-native applications. Delivered models with up to 95% accuracy and automation systems improving efficiency by 30–40%. Passionate about learning, collaboration, and building scalable, secure, intelligent systems.



TECHNICAL SKILLS

Programming	: Python, C++, C, SQL, Bash, PowerShell, Shell Scripting
AI/ML & Generative AI	: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, LLMs, RAG, LangChain, LangGraph, Prompt Engineering, AI Agents, Fine-Tuning, Vector Databases
Frameworks & Libraries	: TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Streamlit, FastAPI
Web Development	: React.js, Next.js, Angular, HTML, CSS, JavaScript, REST APIs
Cloud & DevOps / MLOps	: AWS, Google Cloud, Firebase, Docker, Kubernetes, CI/CD, GitHub Actions, ML Pipelines, Model Deployment, Model Monitoring
Databases	: MySQL, PostgreSQL, MongoDB, Firestore
Core CS	: DSA, OOP, DBMS, Operating Systems, IoT, AINN
Security & DevSecOps	: Static Code Analysis, Dynamic Code Scanning, Vulnerability Assessment, Secure API Integration, OWASP Top 10



EXPERIENCE

AI and ML – Krishi Vigyan Kendra (KVK)

Jan 2026 – May 2026

Baramati

AI/ML Intern

- Built YOLO-based sugarcane disease detection and smart irrigation system using Roboflow annotation, computer vision, predictive analytics, React full-stack development, database integration, and cloud services (AWS, Firebase).
- Achieved 92% detection accuracy and automated irrigation workflows, improving water efficiency by 30%.

Artificial Intelligence and Machine Learning Intern – Uptoskill

Jan 2026 – May 2026

Remote

AI/ML Intern

- Developed LLM, RAG, AI Agent, surveillance, and anomaly monitoring systems using LangChain, LangGraph, YOLO, OpenCV, Streamlit, FastAPI, and Cloud APIs.
- Optimized pipelines and caching strategies, improving retrieval speed by 35% and real-time inference performance by 40%.



PROJECTS

AgriAura – AI-Powered Smart Irrigation Ecosystem

Tech: Python, ML, IoT, ESP8266, Firebase, OpenWeather API, NDVI, AWS

- Built a real-time predictive irrigation platform using IoT sensor streams, weather intelligence, and ML models.
- Achieved **30% water savings** and **45% higher irrigation accuracy**.
- Copyright registered by Govt. of India (LD-32050/2025-CO) and certified by Bharat Forge.

VisionGuard – AI Obstacle Awareness System

Tech: Raspberry Pi, YOLO, OpenCV, Depth Sensor, Ultrasonic, Python

- Engineered a YOLO-powered edge-AI obstacle detection system using deep learning and sensor fusion.
- Achieved **90% detection accuracy** and improved navigation assistance by **40%** for visually impaired users.
- Copyright registered by Govt. of India (LD-46387/2025-CO)

Vital Monitoring & Predictive Healthcare System

Tech: Python, ML, IoT, ESP32, Cloud, Streamlit, Data Analytics

- Built real-time cardiac risk prediction using Python, ML, IoT (ESP32), Streamlit, Cloud, and analytics.
- Achieved **92% accuracy** with anomaly detection and automated cloud alerts.
- Certified by Bharat Forge.

CO₂ Emission & Failure Prediction System

Tech: Python, ML, ESP32, Sensor Fusion, Anomaly Detection, AWS

- Built ML-driven anomaly detection and predictive maintenance platform using Python, ML, ESP32, sensor fusion, AWS with real-time CO₂ emission analytics.
- Achieved **95% fault detection accuracy** and reduced unplanned downtime by **28%**.



EDUCATION

B. TECH-ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology
2023 - 2027 | SGPA:8.43/10

HONOR-CLOUD COMPUTING AND VIRTUALIZATION

Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology
2024 - 2027 | SGPA:9.25/10

HIGHER SCHOOL CERTIFICATE (HSC)

MSBSHSE Completed 2022 | 82.83%

HIGHER SCHOOL CERTIFICATE (HSC)

MSBSHSE Completed 2020 | 93.80%



ACHIEVEMENTS & CERTIFICATIONS



Hackathon & Technothon Achievements

Led Team AIMERS to 7+ Hackathon & Technothon wins including Technothon, Navdhara, Voltzfest, and Elysium by developing AI, Computer Vision, IoT, and predictive systems, earning 2 Govt. of India Copyright Registrations.



Community Leadership

Served as GDG Core Member, IEEE Member, SAAI Association Treasurer, and Bharat Forge Volunteer, mentoring 100+ students in AI, IoT, and Embedded Systems workshops.



Certifications & Technical Credentials

Certified in Generative AI, Cloud Computing, Embedded AI, Full-Stack Development, AR/VR, AI Agents, and MLOps. Bharat Forge-recognized AI projects.

[View All Certificates & Technical Credentials](#)